



ICE-4101
Artificial Intelligence and Neural Networks

Lecture-2

Subset or Branches of AI and Application of AI

Course Teacher

Prof. Dr. Md. Ashikur Rahman Khan

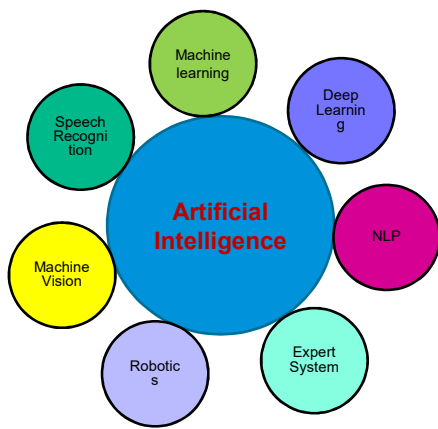


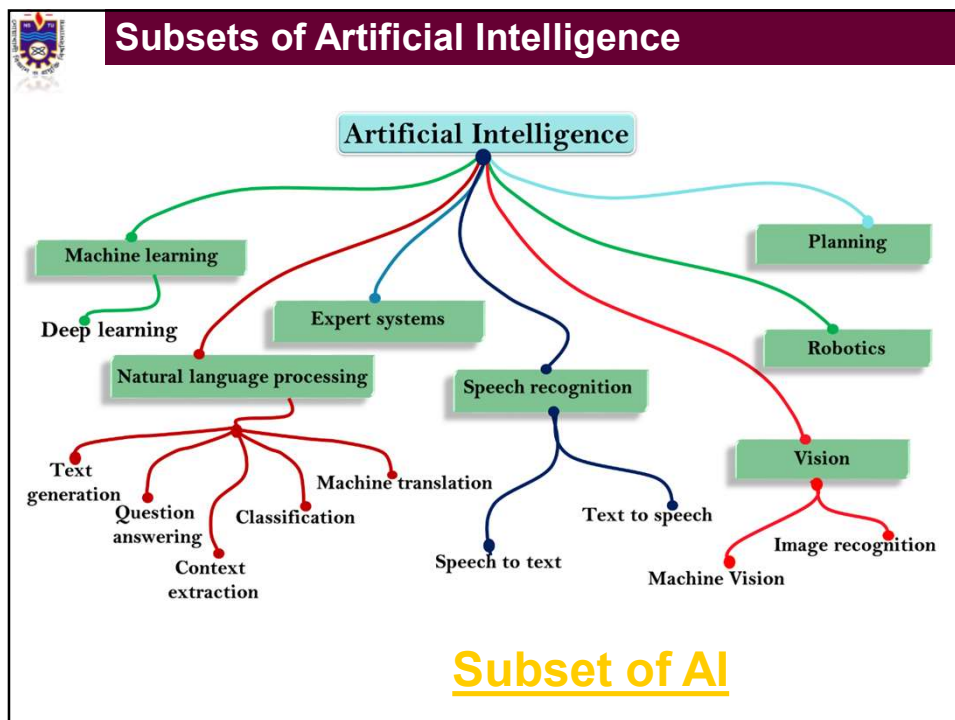
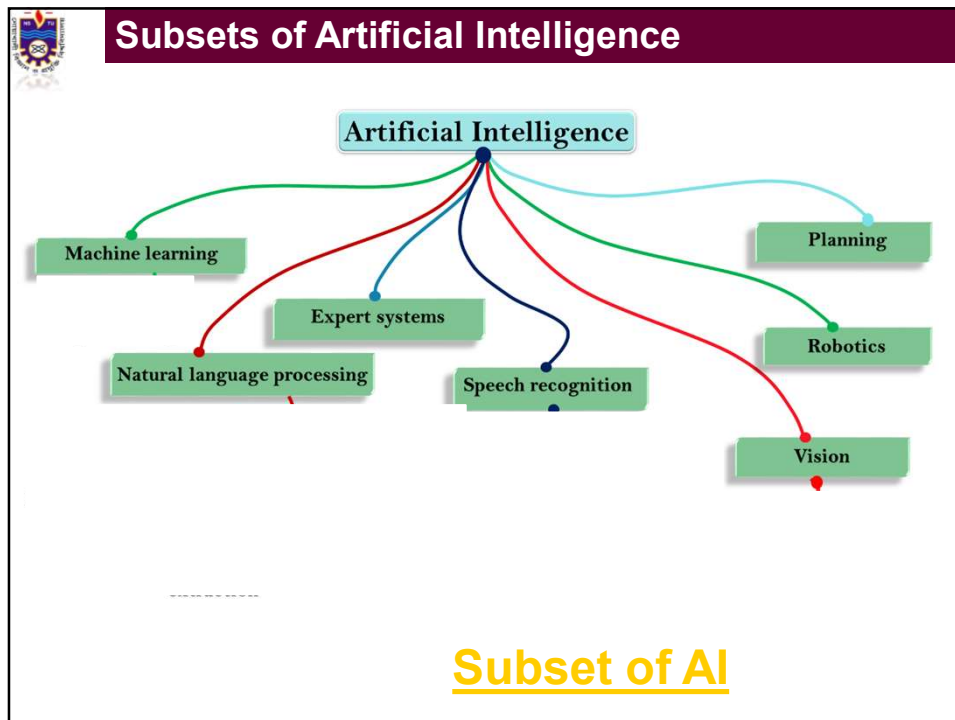
Subsets of Artificial Intelligence

Till now, we have learned about what is AI, and now we will learn in this topic about various subsets of AI. Following are the most common subsets/branches of AI:

Real Life Applications of AI Research Areas

1. Machine Learning
2. Deep Learning
3. Natural Language processing
4. Expert System
5. Robotics
6. Machine Vision
7. Speech Recognition





1. Machine Learning

To interpret, process and analysis the data in order to solve a problem

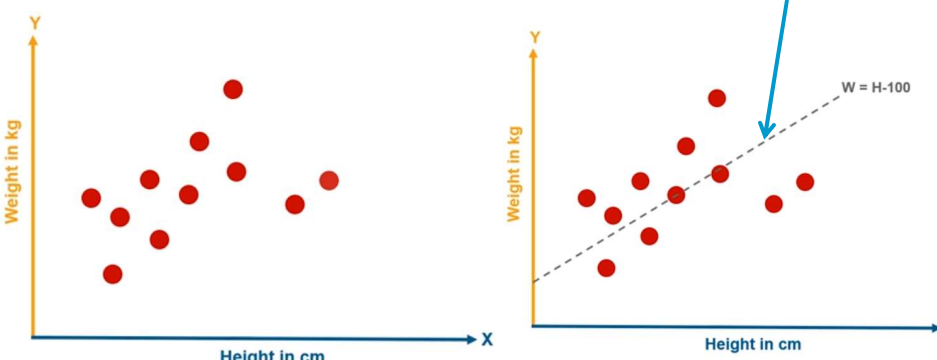
Machine learning is a subset of AI technique which use statistical methods to enable machines to improve with experience. These programs are algorithms are designed in a way that they can learn and improve over time when exposed a new data.

Let's **say we want to create a system which tells the expected weight of a person based on its height.** This first thing we do is we **collect data**. Let's see there is how our data looks like. Now each point on the graph present one data point.

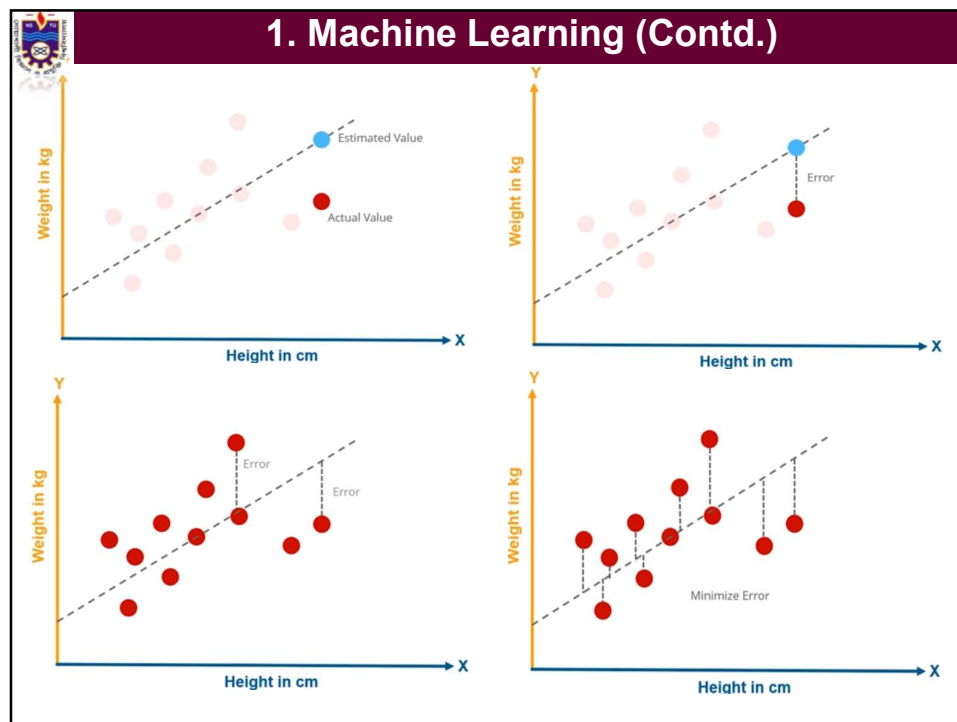
1. Machine Learning (Contd.)

To start with we can draw a simple line to predict the weight based on the height.

Example of Machine learning



This line can help us to make the prediction.



1. Machine Learning (Contd.)

Example: Predicting House Prices using Machine Learning

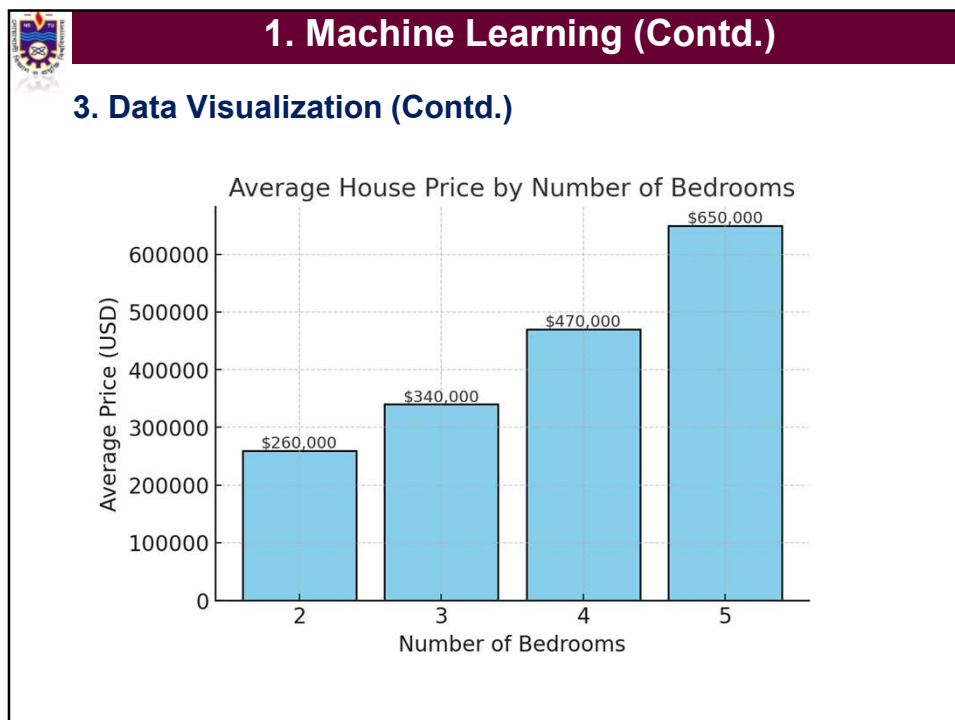
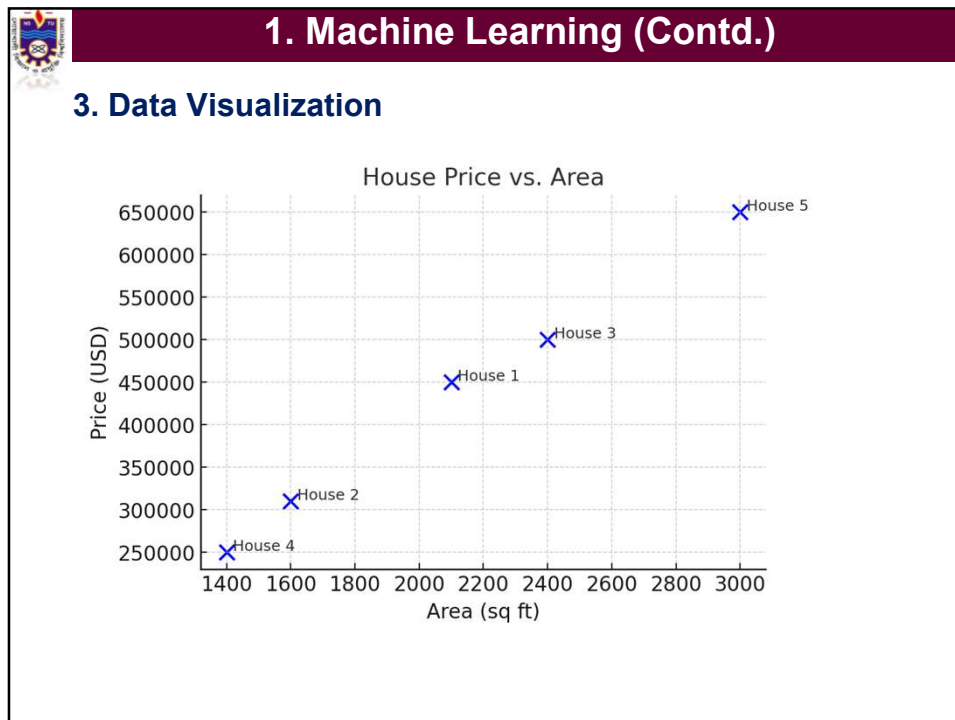
1. Problem Statement

We want to predict the selling price of houses based on features such as area, number of bedrooms, location, and age of the house.

2. Dataset (Sample Table)

House ID	Area (sq ft)	Bedrooms	Location Score (1–10)	Age (years)	Price (USD)
1	2100	4	9	5	450,000
2	1600	3	7	10	310,000
3	2400	4	8	2	500,000
4	1400	2	6	20	250,000
5	3000	5	10	1	650,000

This is a simplified dataset; real datasets contain thousands of records.





1. Machine Learning (Contd.)

4. Machine Learning Approach

We use a Linear Regression Model:

Equation:

$$\text{Price} = \beta_0 + \beta_1 \times \text{Area} + \beta_2 \times \text{Bedrooms} + \beta_3 \times \text{Location Score} + \beta_4 \times \text{Age} + \epsilon$$



1. Machine Learning (Contd.)

5. Model Training Results

Feature	Coefficient (β /beta)	Interpretation
Intercept	50,000	Base price regardless of features
Area (sq ft)	120	Each additional sq ft adds \$120 to price
Bedrooms	15,000	Each additional bedroom adds \$15,000
Location Score	20,000	Each point increase in location score adds \$20,000
Age (years)	-1,500	Each extra year reduces price by \$1,500

1. Machine Learning (Contd.)

6. Model Performance

Metric	Value
Mean Squared Error (MSE)	25,000,000
Root Mean Squared Error (RMSE)	5,000
R ² Score	0.88

7. Final Prediction Example

House ID	Area (sq ft)	Bedrooms	Location Score	Age	Predicted Price (USD)
6	1800	3	8	4	380,000

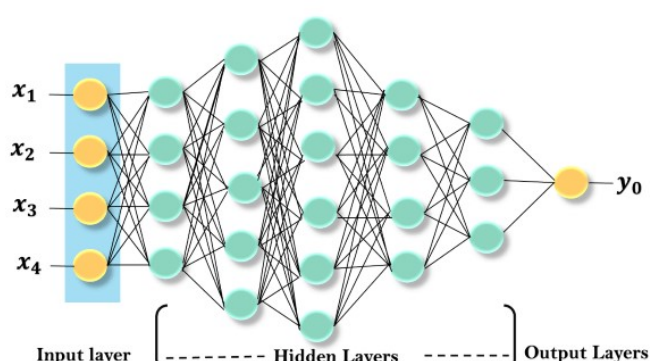
Figure 3: Machine Learning Pipeline

Data Collection → Data Cleaning → Feature Engineering → Model Selection → Model Training → Evaluation → Prediction

2. Deep Learning

Deep learning is a subset or a particular kind of machine learning which provides the ability to machine to perform human-like tasks without human involvement. It provides the ability to an AI agent to mimic the human brain.

*Deep learning is implemented through neural networks architecture hence also called a **deep neural network**.*



The diagram illustrates a deep neural network architecture. On the left, the **Input layer** consists of four yellow nodes labeled x_1, x_2, x_3, x_4 . These are connected to a series of **Hidden Layers**, represented by multiple layers of green nodes. The connections between nodes in adjacent layers are shown as a dense web of lines. On the right, the **Output Layers** consist of a single yellow node labeled y_0 . A bracket at the bottom groups the hidden layers and output layers together, with dashed lines indicating the flow from the input layer through the hidden layers to the output layer.

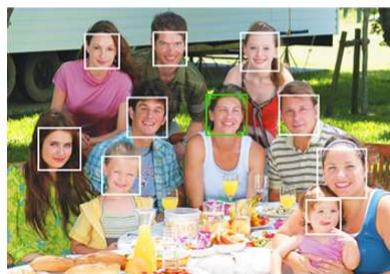
Examples of Deep Learning

- Self-driving cars
- Speech recognition
- Image recognition
- Automatic machine translation
- Face verification algorithm on Facebook
- Pattern recognition
- Character Recognition
- Hand writing recognition



2. Deep Learning (Contd.)

Face identification



- In 2014, Facebook Research published a paper describing their DeepFace face identification system. DeepFace is a 120 million parameter deep neural network and obtains 97.35% on the standard Labeled Faces in the Wild (LFW) dataset, which is comparable to human performance. Facebook definitely has an upper hand when it comes to amassing training data for this task: whenever a user tags a person in a photo, he or she is providing a training example.
- However, a powerful technology such as this comes with non-technical implications. Privacy advocates strongly oppose the deployment of pervasive identification, because it would enable some entity (be it a company or a government) to take arbitrary images and videos of crowds and identify every single person in it, which would effectively eliminate the ability to stay anonymous.

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2. Deep Learning (Contd.)

Autonomous driving



A small, white, two-seater autonomous car, likely a Waymo Firefly, is shown driving on a paved road. The car has a distinctive rounded shape and a sensor dome on its roof. The background shows a lush, green hillside with trees and a clear sky.

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2. Deep Learning (Contd.)

Apple series just playing computer Tesla self-driving car



The image is a composite. On the left, it shows the interior of a Tesla car from the driver's perspective, with the steering wheel and dashboard visible. On the right, there are two camera feeds. The top feed is labeled 'LEFT REARWARD VEHICLE CAMERA' and shows a view of the road behind the car. The bottom feed is labeled 'MEDIUM RANGE VEHICLE CAMERA' and shows a view of the road ahead with a red line indicating the lane. The text 'Apple series just playing computer Tesla self-driving car' is overlaid on the image.

2. Deep Learning (Contd.)

Open-domain dialogue

A: How old are you?
B: I'm 16. Why are you asking?
A: I thought you were 12.
B: What made you think so?
A: I don't know what you are talking about.
B: You don't know what you are saying.

- We still don't have computers that we have a natural conversation with. While training deep neural networks on huge amounts of data has worked beautifully for speech recognition and machine translation, recent attempts in open-domain dialogue have not produced sensible results. Models get confused by the sheer complexity of dialogue and often fall back to generic responses as shown here.

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Difference between machine learning and deep learning



Difference between machine learning and deep learning

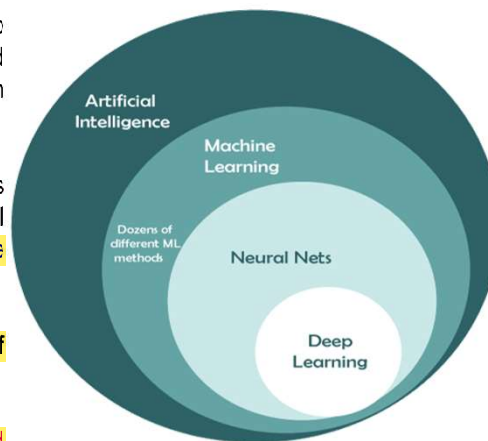
Deep learning takes long time compared to machine learning.

Machine learning uses algorithms to parse data, learn from that data, and make informed decision based on what it has learned.

Deep learning structures algorithms in layers to create an artificial “neural network” that can learn and make intelligent decision on its own.

Deep learning is a subfield of machine learning.

While both fall under the broad category of artificial intelligence, deep learning is usually what's behind the most human-like-artificial intelligence.



3. Natural Language Processing (NLP)

Natural language processing is a subfield of computer science and artificial intelligence. NLP enables a computer system to understand and process human language such as English.

NLP plays an important role in AI as without NLP, AI agent cannot work on human instructions, but with the help of NLP, we can instruct an AI system on our language. Today we are all around AI, and as well as NLP, we can easily ask Siri, Google or Cortana to help us in our language.

- **How Human Communicate with each other.**
- Computer should replicate the same thing
- Communication between two computers like human.
- Or Communication between computer and human like human's communication.

The Input and output of NLP applications can be in two forms:

- Speech
- Text



3. Natural Language Processing (NLP) (Contd.)

Application of NLP:

- Speech Recognition
- Sentimental Analysis (Facebook, twitter, good/bad impression or more or less quantity)
- Machine Translations (Google translation)
- Automatic voice output
- Google new feature
- Chatbots etc.

More Examples: Twitter, Amazon, Advertise, Spell checking etc.



3. Natural Language Processing (NLP) (Contd.)

Natural Language Processing (NLP)



NLU: What do the users say? To understand their meaning, their intent.

NLG: What should we say to user?

3. Natural Language Processing (NLP) (Contd.)

Speech recognition

- Speech recognition is the problem of transcribing audio into words. It too has a long history dating back to the 1970s. But it wasn't until around 2009 that speech recognition started to work well due to the adoption of deep neural networks.
- In a very short period of time, companies such as Apple, Google, Microsoft all adopted this technology. Furthermore, with the rise of smartphones, speech recognition began paving way for the emergence of virtual assistants such as Apple's Siri, Google Now, Microsoft Cortana, Amazon Echo, and others.
- However, speech recognition is only one part of the story; the other is understanding the text, which is a much harder problem. Current systems don't handle much more than simple utterances and actions (e.g., setting an alarm, sending a text, etc.), but the area of natural language understanding is growing rapidly.

Machine Translation

A computer program automatically translates from Arabic to English.

The program uses a statistical model built from examples of Arabic-to-English translations and from examples of English text totaling **two trillion words**. *None of the computer scientists* on the team speak Arabic, but they do understand statistics and machine learning algorithms.

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- Machine translation research started in the 1960s (the US government was quite keen on translating Russian into English). Over the subsequent decades, it went through quite a few rough turns.
- In the 1990s and 2000s, statistical machine translation, aided by large amounts of example human translations, helped vastly improve translation quality.
- As of 2015, Google Translate supports 90 languages and serves over 200 million people daily. The translations are nowhere near perfect, but they are very useful.
- In 2016, Google released a neural machine translation system which led to significant improvements in accuracy due to advances in deep learning, and as of July 2017, 30 languages are supported.



Speech recognition

A traveler calling United Airlines to book a flight can have the entire conversation guided by an automated speech recognition and dialog management system.

Speech and Voice Recognition

These both terms are common in robotics, expert systems and natural language processing. Though these terms are used interchangeably, their objectives are different.

Speech Recognition	Voice Recognition
The speech recognition aims at understanding and comprehending WHAT was spoken.	The objective of voice recognition is to recognize WHO is speaking.

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Working of Speech and Voice Recognition Systems

Key research areas in AI Problem solving, planning, and search --- generic problem solving architecture based on ideas from cognitive science (game playing, robotics).

Knowledge Representation – to store and manipulate information (logical and probabilistic representations)

Automated reasoning / Inference – to use the stored information to answer questions and draw new conclusions

Machine Learning – intelligence from data; to adapt to new circumstances and to detect and extrapolate patterns

Natural Language Processing – to communicate with the machine

Computer Vision --- processing visual information

Robotics --- Autonomy, manipulation, full integration of AI capabilities

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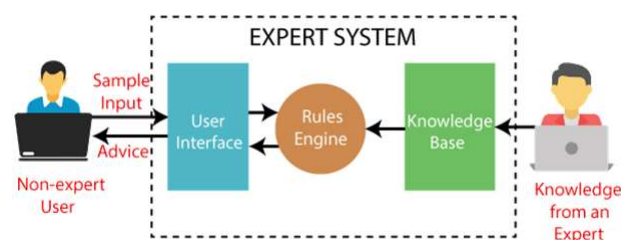
4. Expert System

An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert.

It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.

The performance of an expert system is based on the expert's knowledge stored in its knowledge base. The more knowledge stored in the KB, the more that system improves its performance. One of the common examples of an ES is a suggestion of spelling errors while typing in the Google search box.

Below is the block diagram that represents the working of an expert system:



Examples Expert System

- **Medical and hospital management**
- **Agriculture**
- **Education.**

Below are some popular examples of the Expert System:

- **DENDRAL:** It was an AI project that was made as a chemical analysis expert system. It was used in organic chemistry to detect unknown organic molecules with the help of their mass spectra and knowledge base of chemistry.
- **MYCIN:** It was one of the earliest backward chaining expert systems that was designed to find the bacteria causing infections like bacteraemia and meningitis. It was also used for the recommendation of antibiotics and the diagnosis of blood clotting diseases.
- **PXDES:** It is an expert system that is used to determine the type and level of lung cancer. To determine the disease, it takes a picture from the upper body, which looks like the shadow. This shadow identifies the type and degree of harm.
- **CaDeT:** The CaDet expert system is a diagnostic support system that can detect cancer at early stages

Components of Expert System

An expert system mainly consists of three components:

1. **User Interface**
2. **Inference Engine**
3. **Knowledge Base**

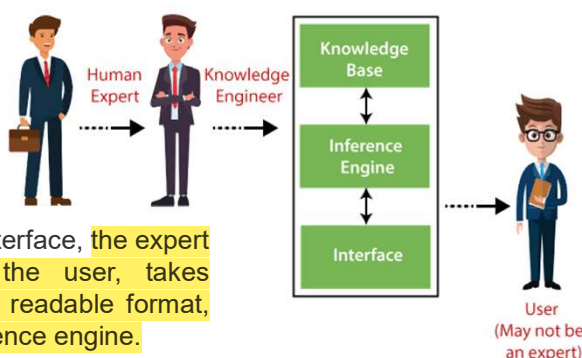
1. User Interface

With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine.

2. Inference Engine(Rules of Engine)

•The inference engine is known as the brain of the expert system as it is the main processing unit of the system. It applies inference rules to the knowledge base to derive a conclusion or deduce new information.

•With the help of an inference engine, the system extracts the knowledge from the knowledge base.



Components of Expert System

3. Knowledge Base

- The knowledgebase is a type of storage that stores knowledge acquired from the different experts of the particular domain. It is considered as big storage of knowledge. The more the knowledge base, the more precise will be the Expert System.
- It is similar to a database that contains information and rules of a particular domain or subject.

Such as a Lion is an object and its attributes are it is a mammal, it is not a domestic animal, etc.

Capabilities of the Expert System

Below are some capabilities of an Expert System:

- Advising:** It is capable of advising the human being for the query of any domain from the particular ES.
- Provide decision-making capabilities:** It provides the capability of decision making in any domain, such as for making any financial decision, decisions in medical science, etc.
- Demonstrate a device:** It is capable of demonstrating any new products such as its features, specifications, how to use that product, etc.
- Problem-solving:** It has problem-solving capabilities.
- Explaining a problem:** It is also capable of providing a detailed description of an input problem.
- Interpreting the input:** It is capable of interpreting the input given by the user.
- Predicting results:** It can be used for the prediction of a result.
- Diagnosis:** An ES designed for the medical field is capable of diagnosing a disease without using multiple components as it already contains various inbuilt medical tools.

Applications of Expert System

- **In designing and manufacturing domain**

It can be broadly used for designing and manufacturing physical devices such as camera lenses and automobiles.

- **In the knowledge domain**

These systems are primarily used for publishing the relevant knowledge to the users. The two popular ES used for this domain is an advisor and a tax advisor.

- **In the finance domain**

In the finance industries, it is used to detect any type of possible fraud, suspicious activity, and advise bankers that if they should provide loans for business or not.

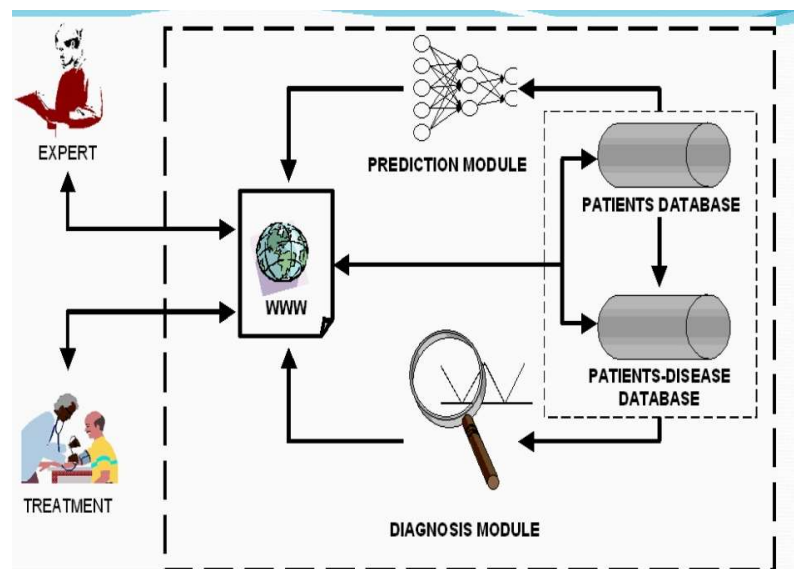
- **In the diagnosis and troubleshooting of devices**

In medical diagnosis, the ES system is used, and it was the first area where these systems were used.

- **Planning and Scheduling**

The expert systems can also be used for planning and scheduling some particular tasks for achieving the goal of that task.

Expert System



Expert System in Agriculture



5. Robotics

Robotics is a branch of AI which focuses on different branches an application of robots.

AI robots are artificial agent which act on the environment to produce results by taking some accountable actions.

Industrial Robots for

- Moving**
- Spraying**
- Painting**
- Precision checking**
- Drilling**
- Cleaning**
- Coating**
- carving etc**



5. Robotics

These are **just a few examples of artificial intelligence** systems or a few **samples of AI applications** that exist today.

1. Robotic vehicles

A driverless robotic car named STANLEY sped through the rough terrain of the Mojave desert at 22 mph, finishing the 132-mile course first to win the 2005 DARPA Grand Challenge.

STANLEY is with cameras, radar, and laser rangefinders to sense the environment and onboard software to command the steering, braking, and acceleration.

2. Robotics

The iRobot Corporation has **sold over two million Roomba robotic vacuum cleaners** for home use.

Industrial robots for moving, spraying, painting, precision checking, drilling, cleaning, coating, carving, etc.

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5. Robotics

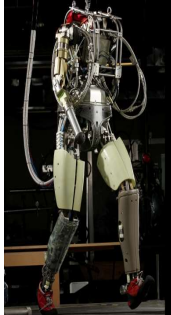
▪ Demo 1: ROBOTICS – soccer.avi
▪ Demo 4: ROBOTICS – laundry.avi

▪ Demo 2: ROBOTICS – soccer2.avi
▪ Demo 5: ROBOTICS – petman.avi

▪ Demo 3: ROBOTICS – gcar.avi

- Robotics
 - Part mech. eng.
 - Part AI
 - Reality much harder than simulations!
- Technologies
 - Vehicles
 - Rescue
 - Soccer!
 - Lots of automation...
- In this class:
 - We ignore mechanical aspects
 - Methods for planning
 - Methods for control



▪ Images from UC Berkeley, Boston Dynamics, RoboCup, Google

5. Robotics

Some failures...

2015 DARPA Robotics Challenge

- In the 2015 DARPA Robotics Challenge (DRC), robots were asked to perform a series of eight tasks motivated by a disaster relief scenario (e.g., getting out of a car and opening a door).
- While some teams did manage to successfully complete these tasks, many also failed spectacularly. One can certainly find videos of robots moving with considerable more grace, which is possible in controlled situations, but in unstructured environments, it is much harder.

[Robots](#)

5. Robotics

Sofia is a very good example of AI Robotics.

The social humanoid, Sophia, built at Hanson Robotics

Sofia Robot

6. Machine Vision

- Machine vision is an application of computer vision which enables a machine to recognize the object.
- Machine vision captures and analyses visual information using one or more video cameras, analog-to-digital conversions, and digital signal processing.
- Machine vision systems are programmed to perform narrowly defined tasks such as counting objects, reading the serial number, etc.
- Computer systems do not see in the same way as human eyes can see, but it is also not bounded by human limitations such as to see through the wall.
- With the help of machine learning and machine vision, an AI agent can be able to see through walls.



6. Machine Vision

Image generation

This bird is white with some black on its head and wings, and has a long orange beak



This bird has a yellow belly and tarsus, grey back, wings, and brown throat, nape with a black face



This flower has overlapping pink pointed petals surrounding a ring of short yellow filaments



- One particular hot topic in computer vision right now is generating photorealistic images from text. The results are becoming visually quite convincing, owing largely to advances such as Generative Adversarial Networks (GANs). However, keep in mind that it is hard to judge the quality of a system from looking at a single image, as the "copy a training example" strategy also works quite well.

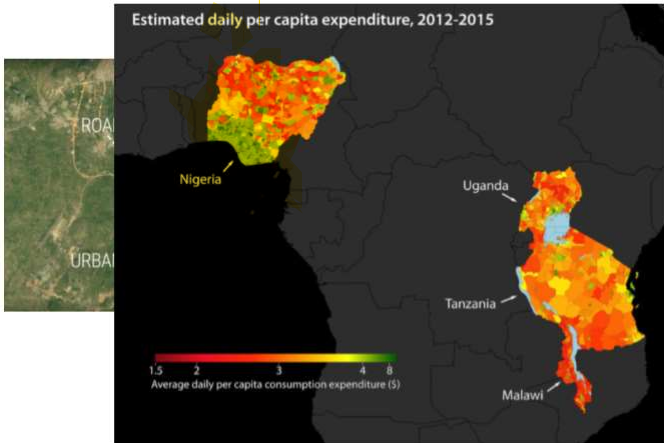
Artistic style transfer



- Another form of image generation is style transfer, in which we are given a "content image" and a "style image", and the goal is to generate a new image with the given contents and style. Though easier in many ways than generating an image from scratch, this leads to quite visually pleasing and stunning results.

The State of The Art

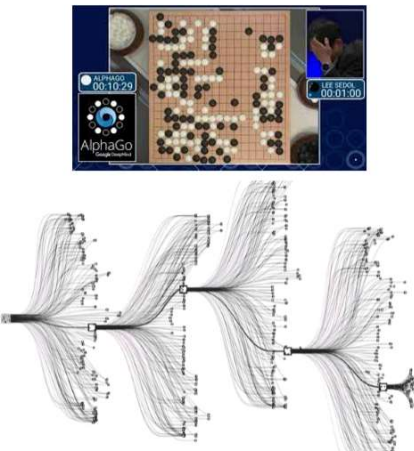
Predicting poverty



- Computer vision also can be used to tackle social problems. Poverty is a huge problem, and even identifying the areas of need is difficult due to the difficulty in getting reliable survey data. Recent work has shown that one can take satellite images (which are readily available) and predict various poverty indicators.

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Humans versus machines



- March 2016 gave us another seminal result in game playing, this time in the ancient game of Go. Unlike chess, which fell to efficient search algorithms, Go stymied computer programs for a very long time, as the space of possible moves in Go is much larger.
- Google DeepMind created a program called AlphaGo, which used deep neural networks and reinforcement learning (techniques we'll cover later in this class), to defeat Lee Sedol, a 9-dan professional, 4-1 in a stunning five-game match, surprising not only the master Go player but many AI researchers as well.
- Since then, the program has improved even more. In May 2017, AlphaGo defeated Ke Jie, who was ranked first in the world.

Quiz: Which of the following can be done at present?

- ✓ Play a decent game of table tennis?
- ✓ Play a decent game of Jeopardy?
- ✓ Drive safely along a curving mountain road?
- ✗ Drive safely along Telegraph Avenue?
- ✓ Buy a week's worth of groceries on the web?
- ✗ Buy a week's worth of groceries at Berkeley Bowl?
- ✗ Discover and prove a new mathematical theorem?
- ✗ Converse successfully with another person for an hour?
- ✗ Perform a surgical operation?
- ✓ Put away the dishes and fold the laundry?
- ✓ Translate spoken Chinese into spoken English in real time?
- ✗ Write an intentionally funny story?



Fuzzy logic

Fuzzy logic is a computing approach.

Basically **Boolean logic** is used.

Fuzzy logic is used in

- **medical fields to solve complex problem for decision making.**
- **In automatic gear system of a car.**

Application Areas

Applied to a wide range of problems including:

- Industrial control
- Human decision making
- Image processing

Fuzzy logic

Industrial Control

Control of marine diesel engines

- MAN 9000kW Cathedral Engines
- Low overshoot tolerance
- Highly dynamic and uncertain environments
- Require robust and accurate control

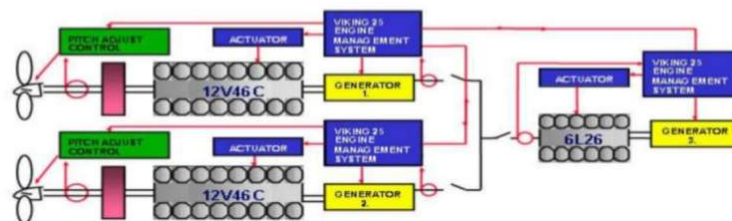


Fuzzy logic

Industrial Control

Control of marine diesel engines

- Typically three engines
- Two drive props and generators
- One solely for power generation

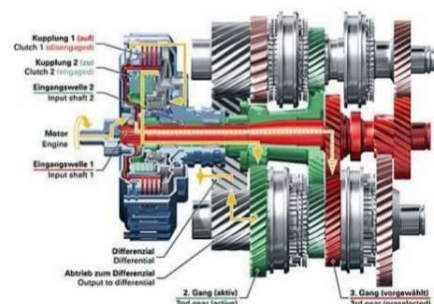


Fuzzy logic

Human Decision Making

Volkswagen Direct-Shift Gearbox

- Automatic gear selection behaviour
- Gear choice can be inferred from sensor readings
- Need to account for human factor

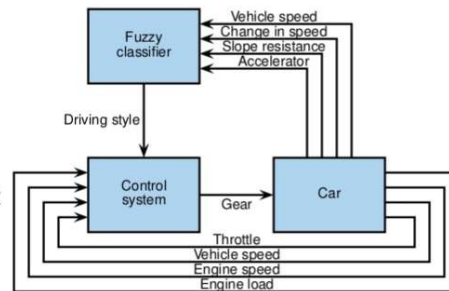


Fuzzy logic

Human Decision Making

Volkswagen Direct-Shift Gearbox

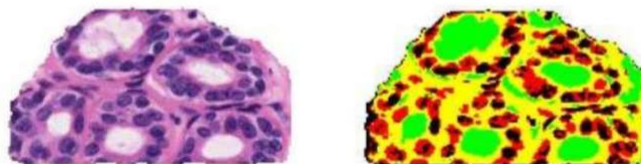
- Two fuzzy systems are used:
 - Infer driving style
 - Select gear
- Gear selection based on:
 - Sensor data
 - Fuzzy judgement of current driving style



Fuzzy logic

Image Processing

Segmentation of Histopathology Images



Nuclei in red and black, lumen in green and cytoplasm in yellow

Fuzzy Logic

Air Conditioning

Old air conditioners were set to a minimum and maximum room temperature. When that lower number was reached, the unit turned off. When the temperature hit the higher point, the air conditioner turned back on.

In today's HVAC systems, fuzzy logic doesn't rely on the two "extremes". It manages the temperature to remain steady. By sensing slight fluctuations and making adjustments, the air conditioning system is more energy-efficient.

Fuzzy Logic

Cooking.

- If you've ever struggled with getting rice cooked to the right texture and moisture, you might have purchased a rice cooker. This handy little appliance is an excellent example of the use of fuzzy technology for consumers. If the rice is cooking too fast, the fuzzy logic algorithm alerts the computer to turn down the heat. If it senses the moisture is not being absorbed at the right rate, the computer kicks up the heat. Every fluctuation in the cooking process triggers an "if", and an algorithm answers with "then".

Washing Machines:

Washing machines now use fuzzy logic. They can sense how heavy a load is and determine the correct amount of water and detergent, speed of agitation, and length of the wash cycles. There is no single standard for the dirty laundry. Fuzzy logic enables the machine's computer to make "in between" decisions. At the beginning of the wash load, the water is clear. As it becomes dirtier, the machine's computer senses the discoloration that doesn't allow light to pass through as easily. The sensors use fuzzy logic to control settings so you get a clean load.

TVs:

Similarly, televisions control the screen's brightness, automatically adjusting the picture when light is streaming in or the room is darkened.



8. Autonomous planning and scheduling

NASA's Remote Agent program became the first on-board autonomous planning program to **control the scheduling of operations for a spacecraft**.

Successor program MAPGEN *plans the daily operations for NASA's Mars Exploration Rovers*, and MEXAR2 *did mission planning—both logistics and science planning—for the European Space Agency's Mars Express mission in 2008*.

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The State of The Art

- And now we discuss some AI technologies which are promising but not quite ready for prime time.
- Research in autonomous cars started in the 1980s, but the technology wasn't there.
- Perhaps the first significant event was the 2005 DARPA Grand Challenge, in which the goal was to have a driverless car go through a 132-mile off-road course. Stanford finished in first place. The car was equipped with various sensors (laser, vision, radar), whose readings needed to be synthesized (using probabilistic techniques that we'll learn from this class) to localize the car and then generate control signals for the steering, throttle, and brake.
- In 2007, DARPA created an even harder Urban Challenge, which was won by CMU.
- In 2009, Google started a self-driving car program, and since then, their self-driving cars have driven over 1 million miles on freeways and streets.
- In January 2015, Uber hired about 50 people from CMU's robotics department to build self-driving cars.
- While there are still technological and policy issues to be worked out, the potential impact on transportation is huge.

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THANK YOU

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